

Unbiased Diffeomorphic Atlas Construction for Computational Anatomy

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Problem formulation

$$\{\hat{h}_i, \hat{I}\} = \operatorname{argmin}_{h, I} \underbrace{\sum_{i=1}^N \underbrace{E(I_i \circ h_i, I)^2}_{\text{Likelihood (Matching)}}}_{\text{Prior (Regularization)}} + \underbrace{\int_0^1 \int_{\Omega} \|Lv_i(x, t)\|^2 dx dt}_{\text{Prior (Regularization)}}$$

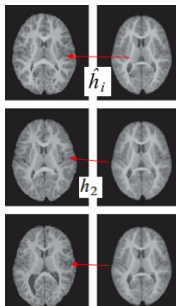


Figure: Atlas template and diffeomorphic transformations

The Core Problem

- Standard atlas construction arbitrarily selects a **single subject** as a template(bias).
- Small-deformations based methods can't handle high anatomical variabilities.

Goals of the Method

- ① **Unbiased Estimation:** Construct a statistical “Mean” independent of any single subject.
- ② **Large Deformations:** Use diffeomorphisms (fluid flows) instead of small-deformation elastic models.
- ③ **Joint Estimation:** Solve for the Template $\hat{I}$ and Transformations h_i simultaneously.

Driving Application

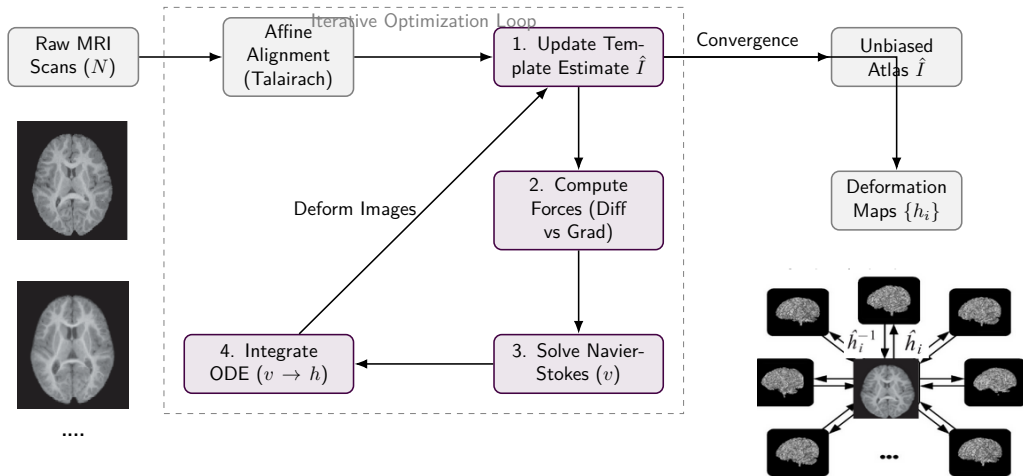
- **Study:** Autism in infants (N=50, age 2 years).
- **Challenge:** High shape variability; lack of normative data for this age group.

Specific Use-Case:

- **Structure:** Caudate Nucleus (weak MRI boundaries).
- **Task:** Construct a probabilistic atlas and segment new subjects via label transfer.



Graphical Abstract



Methodology: The Transformation Model

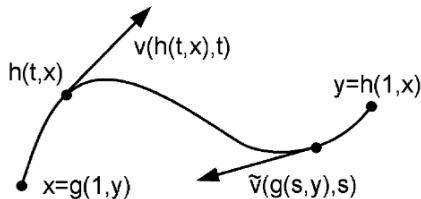
The method moves beyond small deformations ($u(x)$) to **large deformation diffeomorphisms** ($h(x)$).

Diffeomorphism construction

This guarantees topological preservation (no tearing or folding).

$$h(x) = x + \int_0^1 v(h(x, t), t) dt \quad (1)$$

Choice of the velocity field To minimize the data-attachment energy , we construct a force deriving from this energy and define a resulting velocity field (in an analog physical situation)



Methodology: The Minimization Problem

We solve for the **Fréchet Mean** by minimizing the total energy with respect to the Template $\hat{I}$ and Transformations h_i :

$$\{\hat{h}_i, \hat{I}\} = \operatorname{argmin}_{h, I} \underbrace{\sum_{i=1}^N \frac{E(I_i \circ h_i, I)^2}{2}}_{\text{Likelihood (Matching)}} + \underbrace{\int_0^1 \int_{\Omega} \|Lv_i(x, t)\|^2 dx dt}_{\text{Prior (Regularization)}}$$

The Greedy Strategy Solution:

❶ **Template Update:** For fixed transformations, the optimal template is simply the arithmetic mean:

$$\hat{I}(x) = \frac{1}{N} \sum_{i=1}^N I_i(h_i(x)) \quad (2)$$

❷ **Force Computation:** $F(x) = -[I_i(h_i(x)) - \hat{I}(x)] \nabla I_i(h_i(x))$

❸ **Minimization:** $J(v) = \frac{1}{2} \langle Lv, v \rangle - \langle F, v \rangle$

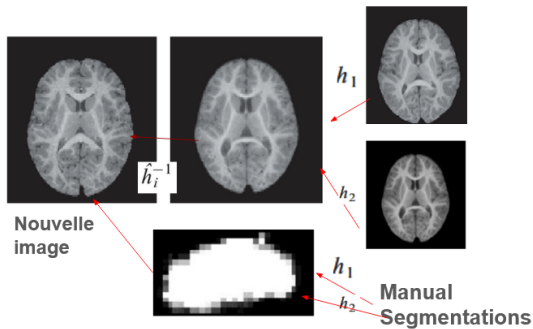
❹ **Velocity Update:** Solve $Lv = F$ (via FFT) and integrate to update h .

Caudate's segmentation algorithm

Require: $\hat{I}$ an atlas template of a set of images with a segmented caudate C_m , I the image with the caudate C we need to segment.

Compute (iteratively) h such that $h(\hat{I}) = I$

Estimate the new segmented caudate $\hat{C}$ by $h(C_m)$



Probabilistic caudate's template

Unbiased Diffeomorphic Atlas

Critical Analysis of Results

Successes

- **Sharpness:** The unbiased atlas retains high-frequency cortical details compared to the blurry affine average.
- **Inverse Consistency:** Results are independent of image ordering.
- **Validation:** Probabilistic Overlap (POV) of **0.85–0.90** with manual experts for Caudate segmentation.

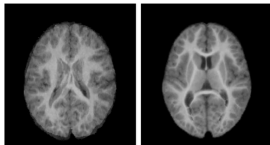


Figure: Arithmetic averaging vs Proposed algorithm

Limitations

- **The “Thin Ventricle” Case:** Subject 5007 showed an **18% volume underestimation**.
- **Cause:** The greedy optimization settled in a **local extremum**.
- **Mechanism:** The energy cost to compress the ventricles (“uphill” deformation energy) outweighed the matching benefit locally.

(1) Ease of Implementation?

It requires:

- Finding the good regularization scale ,and ϵ .
- 3D Fast Fourier Transforms (FFT) for the operator L^{-1} .
- Massive memory management for N dense 3D velocity fields.

(2) Applicability to Other Data?

The framework is generic. Many parameters are not specified, and should be adapted empirically: the regularization term scale, the choice of ϵ .

- Sensitive to contrast change
- Can be used for slow motion tracking (Cardiac motion)
- The dissimilarity metric can be adapted in case of multimodal acquisition

(3) Most Difficult Concept?

Why the algorithm falls into a local minimum.

Why the iterations minimize the objective function.